FINANCIAL MANAGEMENT

Suggested Answers
March-April 2025

Answer to the Question# 1(a):

AD's board must consider **ethical, legal, and business risks** before continuing with this manufacturer. Child labor is widely condemned, and working with unethical suppliers could harm AD's **brand reputation** and lead to **consumer boycotts**. There are also **legal risks**, as international labor laws prohibit child labor. While cutting ties may disrupt production, finding an **ethical supplier** could strengthen AD's commitment to responsible business. Alternatively, AD could require the manufacturer to **implement fair labor practices**. Prioritizing ethical manufacturing helps build long-term trust with customers.

Answer to the Question# 1(b):

The belief that earnings per share (EPS) growth is the best measure of shareholder wealth has limitations. While EPS reflects profitability, it does not capture key value-creation factors, such as cash flow, risk, and long-term sustainability. EPS can also be manipulated through short-term financial strategies like share buybacks, without truly improving business performance.

Instead, businesses should focus on three key value drivers to increase shareholder wealth:

1. Revenue Growth: Expanding market share, enhancing customer experience, and diversifying products boost revenue, leading to sustainable profitability.
2. Operational Efficiency: Reducing costs through better resource utilization and innovation improves profit margins, positively impacting overall financial health.
3. Return on Capital Employed (ROCE): Efficiently using invested capital generates higher returns, ensuring long-term value creation for shareholders.

Managing these drivers fosters real value growth, rather than relying solely on EPS as a measure of financial success.

Answer to the Question# 1(c) (i):

		Each shareholder	Company as a whole	
Value of existing shares	3	at Tk. 2 = Tk. 6	150,000 at Tk. 2 =	Tk. 300,000
Value of capital injected by right issue	1	at Tk. 1 = Tk. 1	50,000 at Tk. 1 =	Tk. 50,000
	4	Shares Tk. 7	200,000	Tk. 350,000

Theoretical ex-rights price = Tk. 7/4 shares (or Tk. 350,000/200,000 shares) = Tk. 1.75 per share

Answer to the Question# 1(c) (ii):

Value of the right to subscribe for each new share
= ex rights price - subscription price
= Tk. 1.75 - Tk. 1
=Tk. 0.75
Value of a right share per existing share = Tk. 0.75/3 = Tk. 0.25

Answer to the Question# 1(d):

To decide between the Sukuk and the Conventional Bond, we should evaluate the risk-return trade-off, income certainty, Shariah compliance, and liquidity. Here's a breakdown:

Option 1: Sukuk
Key Features:

- Profit rate: 7% (projected, not guaranteed)
- Underlying asset: Ownership in income-generating real estate
- Income: Periodic profit distributions based on rental income
- Shariah-compliant

Pros:
Higher projected return (7% vs. 6%)
Asset-backed security — real estate reduces default risk to some extent
Shariah-compliant, suitable for Islamic investors
Potential for capital appreciation depending on asset structure

Cons:
Return is not guaranteed – depends on performance of the underlying properties
Profit distributions may fluctuate if rental income varies
Liquidity may be lower compared to conventional bonds
May carry higher management/structuring costs

Option 2: Conventional Bond

Key Features:

- Interest rate: 6% fixed per year
- Issuer: Established company
- Income: Fixed annual interest payments

Pros:
Predictable income stream — 6% fixed for 5 years
Generally more liquid, especially if listed
Lower volatility — payments not linked to market performance
Simple and widely understood financial instrument
Cons:
Lower return compared to Sukuk (6% vs. 7%)
Interest income is taxable (depending on your jurisdiction)
Not Shariah-compliant
Credit risk — depends on issuer's solvency

Answer to the Question# 2(a):

Weighted Average Cost of Capital (WACC): (E/V * Ke) + {(D/V * Kd) (1-t)}

Where,
Market value of equity (E) = BDT 30 million
Market value of debt (D) = BDT 20 million
Total capital (V) = BDT 50 million
Cost of equity (Ke) = 12%
Cost of debt (Kd) = 8%
Corporate tax rate (t) = 30%

Current WACC = (30/50 *12%) + {(20/50*8%) (1-.30)}
= 9.44%

Option 1: Issuing BDT 20 million in new equity

New Equity Value: BDT 50 million
New Debt Value: BDT 20 million (unchanged)
Total Value = BDT 70 million
WACC with Equity Financing = (50/70 *12%) + {(20/70*8%) (1-.30)}
= 10.17%

Option 2: Taking on BDT 20 million in new long-term debt
New Equity Value: BDT 30 million
New Debt Value: BDT 40 million
Total Value = BDT 70 million
New Cost of Debt = 9%
WACC with Debt Financing = (30/70 *12%) + {(40/70*9%) (1-.30)}

= 8.74%

Option 3: Hybrid Approach (BDT 10 million Equity + BDT 10 million Debt)
New Equity Value: BDT 40 million
New Debt Value: BDT 30 million
Total Value (V) = BDT 70 million
WACC with Hybrid Financing = (40/70 *12%) + {(30/70*9%) (1-.30)}
= 9.56%

Note:
The calculation above is only an indication. The actual result will depend up the new cost of debt and equity under each of the scenarios since the financial risk changes under each of the options 1 to 3 that will either increase or decrease the cost of debt and equity in inverse proportion.

Evaluating the Impact on Company Valuation
WACC represents the company's minimum return requirement for investments to create value. A lower WACC suggests cheaper financing costs and higher valuation potential, whereas a higher WACC implies increased investor expectations. Debt financing results in the lowest WACC (8.74%), meaning lower required returns from investors and potentially higher firm valuation. However, it increases financial risk. Meanwhile, equity financing increases WACC to 10.17%, reducing firm valuation but ensuring financial stability. The hybrid model (9.56% WACC) strikes a balance, optimizing cost while mitigating excessive risk.

Answer to the Question# 2(b):

Expected Earnings Per Share (EPS) under each financing option:

Net Income=EBIT−Interest Expense−Taxes
Since EBIT is expected to grow by 25%:
Post-expansion EBIT=12 × (1.25) = BDT 15 million

Interest Expenses for Each Option:

Option 1 (Equity Financing): No additional debt
Interest Expense remains BDT 1.6 million (8% of existing BDT 20 million)

Option 2 (Debt Financing): New BDT 20 million debt at 9%
Total Interest Expense = BDT 1.6 million + (20 million × 9%) = BDT 3.4 million

Option 3 (Hybrid Financing): New BDT 10 million debt at 9%
Total Interest Expense = BDT 1.6 million + (10 million × 9%) = BDT 2.5 million

Taxable Income and Taxes:

Financing Option	EBIT (BDT million)	Interest Expense (BDT million)	Taxable Income (BDT million)	Taxes (BDT million) @ 30%	Net Income (BDT million)
Option 1 (Equity)	15	1.6	13.4	4.02	9.38
Option 2 (Debt)	15	3.4	11.6	3.48	8.12
Option 3 (Hybrid)	15	2.5	12.5	3.75	8.75

Earnings Per Share: Net Income/ Number of Shares Outstanding

Since the question does not provide a specific share price, I have assumed a BDT 1 per share value for simplicity. Based on this assumption:
Shares Outstanding=Market Value of Equity / Assumed Share Price (BDT 1)
Thus, the initial number of shares before expansion is: BDT 30 million / 1= 30 million shares

For each financing option:

Option 1 (Equity Financing): New equity issuance of BDT 20 million increases total market value to BDT 50 million, meaning estimated shares grow to 50 million.

Option 2 (Debt Financing): No new shares issued, thus remains at 30 million shares.

Option 3 (Hybrid Financing): New equity issuance of BDT 10 million increases total market value to BDT 40 million, meaning estimated shares grow to 40 million.

Financing Option	Net Income (BDT million)	Shares Outstanding	EPS (BDT)
Option 1 (Equity)	9.38	50 million	0.1876
Option 2 (Debt)	8.12	30 million	0.2707
Option 3 (Hybrid)	8.75	40 million	0.2188

Debt Financing (Option 2) results in the highest EPS (BDT 0.2707) because no additional shares were issued, preserving shareholder value. Hybrid Financing (Option 3) has a moderate EPS of BDT 0.2188, balancing risk and dilution. Equity Financing (Option 1) results in the lowest EPS (BDT 0.1876) due to the dilution from issuing new shares.

Answer to the Question# 2(c):

Nova Tech Solutions should consider Option 3 (Hybrid Financing), raising BDT 10 million through equity and BDT 10 million through debt. This approach balances financial risk, cost of capital, and shareholder interests effectively.

Debt financing (Option 2) maximizes Earnings Per Share (EPS) but significantly increases financial risk due to higher interest expenses and leverage. While full equity financing (Option 1) eliminates debt risk, it heavily dilutes shareholder value, resulting in lower EPS.

By choosing a mix of debt and equity, Nova Tech maintains ownership control while limiting debt exposure. The Weighted Average Cost of Capital (WACC) under Option 3 remains lower than full equity financing while avoiding excessive financial burden from debt. This provides long-term flexibility, ensuring the company can sustain growth without over-leverage concerns.

Key risks, such as financial distress due to high debt or shareholder dilution, are moderated under this hybrid model. With controlled debt servicing and partial dilution, Nova Tech can expand efficiently while maintaining strong financial health.

Answer to the Question# 3(a):

Option 1 – Through foreign exchange market:

2 month forward contract with the banker, to sell ¥ 30,000,000 proceeds from Japan
With the forward contract, the company can sell ¥ 30,000,000 at the bank's buying rate of ¥ 1.3755 per BDT.

BDT proceeds under forward contract = ¥30,000,000 * 1.3755
= BDT 41,265,000

Rupee proceeds under future spot rate (bankers buying rate):

Current Spot Rate BDT/¥ = 1.4185

In two months time, the expected spot rate = 1.4185 * 0 .94 = 1.3334

BDT proceeds at the spot rate in 2 moths time = ¥ 30,000,000 * 1.3334
= BDT 40,002,000

Option 2 – Through money market hedging

The company can borrow ¥ 30,000,000 at the rate of 7.5% for two months and convert it into Taka at today's spot rate and then invest the Taka proceeds at the rate of 12% for two months. Finally, the company can settle the yen loan from the yen proceeds in two months' time.

Amount that could be borrowed in yen = ¥30,000,000/(1.0125)
= ¥ 29,629,630

BDT proceeds from conversion = ¥29,629,630 * (1.4185)
= BDT 42,029,630

Investing the Taka proceeds in the money market = BDT 42,029,630 * (1.02)
= BDT 42,870,223

= BDT 2,868,223

The company should go for money market hedging as the proceeds from this option is greater than that arising from using a forward contract.

Answer to the Question# 3(b):

Three Types of Currency Risk (Exposure)

1. **Transaction Exposure**
   - **Definition**: Arises from actual transactions denominated in foreign currencies (e.g., payables, receivables, borrowings).
   - **Example**: The Bangladeshi company is exposed to transaction risk due to its ¥30 million invoice receivable – changes in the Yen/BDT rate will affect the BDT value received.
   - **Hedging**: Forward contracts, money market hedges.
2. **Translation Exposure *(a.k.a. Accounting Exposure)***
   - **Definition**: Occurs when consolidating financial statements of foreign operations that use different currencies.
   - **Example**: A Bangladeshi company with a subsidiary in Japan must convert the subsidiary's Yen-denominated balance sheet into BDT. FX rate changes will affect reported assets, liabilities, and earnings.
   - **Hedging**: Often not hedged actively; use of functional currency strategies or balance sheet hedging.
3. **Economic (Operating) Exposure**
   - **Definition**: Long-term effect of currency fluctuations on a company's **market value** and **competitive position**, affecting future cash flows.
   - **Example**: If Yen appreciates permanently, Bangladeshi software exporters may become relatively more expensive to Japanese clients, losing market share to local or cheaper alternatives.
   - **Hedging**: Strategic—includes diversification, pricing strategies, sourcing adjustments, and relocating production.

Answer to the Question# 4(a):

Forward Rate calculation:
Spot rate = 115 BDT/EUR
Premium = 0.7%.
Forward rate = 115 + (115 × 0.007)
= 115.805 BDT/EUR.

Hedging €100,000:
Receipt = €100,000 × 115.805
= BDT 11,580,500

Unhedged €50,000:
Best-Case Scenario (1 EUR = 117 BDT): €50,000 × 117 = BDT 5,850,000
Total Net Receipt = 11,580,500 + 5,850,000 = BDT 17,430,500

Worst-Case Scenario (1 EUR = 112 BDT): €50,000 × 112 = BDT 5,600,000
Total Net Receipt = 11,580,500 + 5,600,000 = BDT 17,180,500

Answer to the Question# 4(b):

Put Option (€100,000):
Exercise price = 114 BDT/EUR
Premium = 0.8%
Premium cost = 114 × 0.008 = 0.912 BDT/EUR
Net rate = 114 − 0.912 = 113.088 BDT/EUR
Amount = €100,000 × 113.088 = BDT 11,308,800

Unhedged €50,000:
Best-Case Scenario (1 EUR = 117 BDT): €50,000 × 117 = BDT 5,850,000
Worst-Case Scenario (1 EUR = 112 BDT): €50,000 × 112 = BDT 5,600,000

Total Net Receipt:

Using Put Option:
Best case: 11,308,800 + 5,850,000 = BDT 17,158,800
Worst case: 11,308,800 + 5,600,000 = BDT 16,908,800

Answer to the Question# 4(c):

Step-by-Step Arbitrage Possibilities

Option A: EUR–BDT Arbitrage (Basic Money Market Hedge)

Steps:

1. **Borrow** present value of €150,000 in EUR today at 5% annual = **1.25% quarterly**.

PV=€150,000/1.0125=€148,148PV

2. **Convert** €148,148 at spot to BDT:

€148,148×115=BDT17,037,020

3. **Invest** BDT 17,037,020 at 7% annual = 1.75% quarterly:

=17,037,020×1.0175=BDT17,334,666

4. **In 3 months**, pay back €150,000 loan from export proceeds.

Net BDT cash = 17,334,666

Compare with **forward rate hedge = 17,370,750**

So, **no arbitrage gain** here — forward is better.

Option B: EUR → GBP → BDT Arbitrage

We'll explore whether routing through **GBP** using interest and FX differences improves BDT conversion.

Steps:

1. **Convert €150,000 to GBP today** at 1.17:

GBP=€150,000/1.17≈GBP128,205

2. **Invest GBP 128,205** at 4% annual = 1% quarterly:

GBP=128,205×1.01=GBP129,487

3. **Convert GBP to BDT in 3 months**:

- Best case: 129,487 × 137 = BDT **17,740,719**

- Worst case: 129,487 × 132 = BDT **17,090,284**

Expected BDT range = **17.09M – 17.74M**

Compare with:

- Forward: **17.37M**
- EUR money market hedge: **17.33M**
- Option hedge (Put): **16.9M – 17.15M**

So, this strategy **beats all others** in the **best case**, and is **still competitive** in the worst.

Conclusion of Arbitrage Assessment

Strategy	Best Case	Worst Case	Notes
Forward Contract	17,370,750	17,370,750	Fixed, no upside
EUR–BDT Money Market	17,334,666	17,334,666	Slightly worse than forward
Options (Put)	17,158,000	16,908,000	Provides upside, costs premium
EUR→GBP→BDT Arbitrage	**17,740,719**	**17,090,284**	Best upside, but exposed to GBP/BDT volatility

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Recommendation (Part d – Integrated Strategy)

Optimal Hybrid Strategy:

1. **Hedge €100,000 via Forward**
    - Guarantees **BDT 11.58M**
    - Removes risk on 2/3 of exposure
2. **Use EUR→GBP→BDT arbitrage for €50,000**
    - Best-case return: **BDT 5.91M**
    - Even worst-case beats options
    - Total expected BDT range: 17.47M–17.73M

Why this mix?

- Forward covers downside with certainty
- GBP route captures potential upside
- Avoids expensive options
- Doesn't overleverage or lock all funds into one strategy

Answer to the Question# 5(a):

Year	DF	Project A		Project B		Project C		Project D	
		Cash Flow	PV	Cash Flow	PV	Cash Flow	PV	Cash Flow	PV
0	1.0000	-4,698,000	-4,698,000	-3,510,000	-3,510,000	-3,420,000	-3,420,000	-3,510,000	-3,510,000
1	0.8850	1,530,000	1,353,982	1,620,000	1,433,628	1,116,000	987,611	1,584,000	1,401,770
2	0.7831	1,530,000	1,198,214	1,692,000	1,325,084	1,116,000	873,992	1,584,000	1,240,504
3	0.6931	1,530,000	1,060,367	1,530,000	1,060,367	1,602,000	1,110,266	1,584,000	1,097,791
4	0.6133	1,530,000	938,378			1,746,000	1,070,854	1,584,000	971,497
5	0.5428	1,530,000	830,423						

NPV			683,364		309,079		622,723		1,201,563

	DF	Project A		Project B		Project C		Project D		Project E	
		Cash Flow	PV	Cash Flow	PV	Cash Flow	PV	Cash Flow	PV	Cash Flow	PV
0	1	(4,698,000.00)	(4,698,000.00)	(3,510,000.00)	(3,510,000.00)	(3,420,000.00)	(3,420,000.00)				
1	0.885	1,530,000.00	1,354,050.00	1,620,000.00	1,433,700.00	1,116,000.00	987,660.00				
2	0.783	1,530,000.00	1,197,990.00	1,692,000.00	1,324,836.00	1,116,000.00	873,828.00				
3	0.693	1,530,000.00	1,060,290.00	1,530,000.00	1,060,290.00	1,602,000.00	1,110,186.00				
4	0.613	1,530,000.00	937,890.00		-	1,746,000.00	1,070,298.00				
5	0.543	1,530,000.00	830,790.00		-		-		-		-
NPV			**683,010.00**		**308,826.00**		**621,972.00**				

(b) Profitability Index (PI) for Each Project

Project	Profitability Index
A	1.1455
B	1.0881
C	1.1821
D	1.3423
E	1.2417

(c) Rankings

By NPV:

1. **Project D**
2. Project E
3. Project A
4. Project C
5. Project B

By PI:

1. **Project D**
2. Project E
3. Project C
4. Project A
5. Project B

(d) Why Rankings Differ:

The **NPV** method measures **absolute value creation**, favoring projects that contribute the highest total surplus above cost. On the other hand, the **Profitability Index (PI)** focuses on **efficiency per taka invested**, which is crucial under capital constraints.

For example:

- **Project C** ranks below A on NPV, but above it on PI. This indicates that while A creates more value in total terms, C is more efficient per unit of investment.
- **Project D** ranks first in both, showing it is both the most valuable and most efficient investment.

In capital rationing scenarios, **PI is often more useful** for portfolio selection, whereas **NPV best reflects overall value addition** to the firm.

Answer to the Question# 6(a):

Step 1: Initial Capital Outlay

Component	**Amount (BDT million)**
Machinery and Equipment	100.00
Working Capital (recovered Y5)	5.00
Regulatory Compliance Cost	2.00
Total Initial Investment	**107.00**

Step 2: Gross Cash Inflows (Before Adjustments)

Year	**Gross Inflow (BDT million)**
1	40.00
2	45.00
3	50.00
4	55.00
5	70.00

Step 3: Inflation Adjustment (3% per year)

Adjusted Inflows (in real terms):

Year	**Real Inflow = Nominal / (1.03)^n**
1	40.00 / 1.000 = **40.00**
2	45.00 / 1.0609 ≈ **42.40**
3	50.00 / 1.0927 ≈ **45.75**
4	55.00 / 1.1255 ≈ **48.89**

Year	Real Inflow = Nominal / (1.03)^n
5	70.00 / 1.1593 ≈ **60.37**

Step 4: Risk Adjustments

- **Year 1** inflow is reduced by **25%** due to regulatory delay
  ⇒ 40.00 × 0.75 = **30.00**
- **Year 2** inflow is reduced by **5%** due to demand drop
  ⇒ 42.40 × 0.95 ≈ **40.28**

Adjusted Cash Inflows after Risk:

Year	Inflow (BDT million)
1	30.00
2	40.28
3	45.75
4	48.89
5	60.37

Step 5: Tax Adjustment (20%)

Apply 20% tax:

Year	After-tax Inflow = Inflow × 0.80
1	30.00 × 0.80 = **24.00**
2	40.28 × 0.80 = **32.23**
3	45.75 × 0.80 = **36.60**
4	48.89 × 0.80 = **39.11**
5	60.37 × 0.80 = **48.30**

Step 6: Operating Costs

Category	Y1	Y2	Y3	Y4	Y5
Operational Costs (3M ↑5%)	3.00	3.15	3.31	3.47	3.64
Marketing (Y1 & Y2)	2.00	2.00	0.00	0.00	0.00
Admin Overhead	3.00	3.00	3.00	3.00	3.00
Utilities (1M ↑4%)	1.00	1.04	1.08	1.13	1.17
Repairs (Y2 onwards ↑4%)	0.00	2.00	2.08	2.16	2.25

Category	Y1	Y2	Y3	Y4	Y5
Total Costs	9.00	11.19	9.47	9.76	10.06

Step 7: Tax Shield from Depreciation

- Depreciation = 100M ÷ 5 = 20M/year
- Tax Shield = 20M × 20% = **4.00M/year**

Step 8: Net Cash Flows (after costs and tax shield)

Net CF = After-tax inflow − (Total Costs−Tax Shield)

Year	After-Tax Inflow	Total Costs	Tax Shield	Net Cash Flow
1	24.00	9.00	4.00	24.00 - (9.00 - 4.00) = **19.00**
2	32.23	11.19	4.00	32.23 - 7.19 = **25.04**
3	36.60	9.47	4.00	36.60 - 5.47 = **31.13**
4	39.11	9.76	4.00	39.11 - 5.76 = **33.35**
5	48.30	10.06	4.00	48.30 - 6.06 + **15.00** (salvage) + **5.00** (WC) = **62.24**

Step 9: Discount Net Cash Flows at 10%

PV=Net CF / $(1+r)^t$

Year	Net Cash Flow	Discount Factor (10%)	Present Value
1	19.00	0.9091	17.27
2	25.04	0.8264	20.71
3	31.13	0.7513	23.39
4	33.35	0.6830	22.77
5	62.24	0.6209	38.67
			122.81

Step 10: Final NPV Calculation

NPV=Total PV of Net Inflows−Initial Investment

NPV=122.81−107.00=BDT15.81million

Answer to (b)(i)

Sensitivity Analysis: 10% Adverse Change in Net Cash Flows

To assess how sensitive the project's NPV is to a **10% decrease in net cash flows**, we'll:

1. Reduce each year's net cash flow by 10%
2. Recalculate the NPV using the same 10% discount rate
3. Compare the new NPV with the original NPV

Step 1: Original Net Cash Flows (from earlier)

Year	Original Net Cash Flow (BDT million)
1	19.00
2	25.04
3	31.13
4	33.35
5	62.24

Step 2: Apply 10% Reduction

Year	Adjusted Net CF = 90% of Original
1	17.10
2	22.54
3	28.02
4	30.01
5	56.02

Step 3: Discount at 10% Rate

Year	Net CF	Discount Factor	PV (BDT million)
1	17.10	0.9091	15.55
2	22.54	0.8264	18.63
3	28.02	0.7513	21.05
4	30.01	0.6830	20.50
5	56.02	0.6209	34.80
			110.53

Step 4: Recalculate NPV

NPV revised = 110.53−107.00=BDT3.53million

Step 5: Sensitivity Impact Summary

Metric	Value (BDT million)
Original NPV	15.81
NPV after 10% CF drop	3.53
Change in NPV	**-12.28**
% Drop in NPV	**~77.7%**

Conclusion:

- The NPV drops by ~77.7% due to a **10% decrease in annual net cash flows**, indicating that the project is **quite sensitive** to revenue underperformance or cost overruns.
- While it still remains **slightly positive**, the **margin of safety is narrow**.
- This underlines the importance of risk management and contingency planning before launch.

Answer to (b)(ii)

1. Real Options Theory (ROT)

What it does:
Treats investment opportunities like **financial options**, giving the firm flexibility to adapt as uncertainty unfolds.

How to apply:

- **Option to delay**: Wait for regulatory approvals or clearer demand signals before full rollout.
- **Option to expand**: If Year 2 demand is strong, invest further to scale production.
- **Option to abandon**: Exit the project early if cumulative cash flows fall below salvage value threshold.

Benefit:
Encourages *flexible, staged commitment* instead of all-or-nothing decisions.

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2. Risk-Adjusted Return Models

What it does:
Adjusts the **discount rate** or expected returns based on a project's risk profile.

How to apply:

- Use a **higher discount rate** for pessimistic cash flow scenarios or for less certain years (e.g., post-Year 3).
- Incorporate **project-specific risk premiums** based on market conditions, regulatory complexity, or technology uncertainty.

Benefit:
Allows more realistic valuation by penalizing riskier cash flows, improving capital allocation under uncertainty.

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3. Monte Carlo Simulation / Scenario Analysis

What it does:
Runs **thousands of simulations** using random values for inputs (sales volume, costs, delays) to generate a **range of possible NPVs**.

How to apply:

- Assign probability distributions to key variables (e.g. demand growth, repair costs, delays).
- Run simulations to see how often NPV > 0 or IRR > required rate.

Benefit:
Provides **risk insights** (e.g., "80% chance NPV stays positive") and identifies **key value drivers** and risk thresholds.

Conclusion:

Using these tools together, Mark Innovations can:

- Build **flexibility into launch and scale decisions**
- Adjust forecasts for **risk realism**
- Understand and communicate **downside risks and upside potential**

This shifts the board's mindset from "should we invest or not?" to "**how can we invest wisely under uncertainty?**" — a crucial edge for a firm pioneering new technologies.

Answer to (c)(i)

Calculate the Internal Rate of Return (IRR)

To calculate the IRR, we find the rate rrr that makes the **NPV = 0**, using the original net cash flows:

Year	Net Cash Flow (BDT million)
0	-107.00
1	19.00
2	25.04
3	31.13
4	33.35
5	62.24

We can estimate IRR using financial tools or interpolation. Since earlier calculations used a 10% discount rate and gave a **positive NPV**, IRR must be **greater than 10%**.

Using interpolation between 10% and 15% (we estimate NPV at 15%), or solving using Excel/financial calculator:

Estimated IRR ≈ 12.9%

Answer to (c)(ii)

Compare with Alternative Investment

Criteria	**EcoCool Pro**	**Civil Engineering Project**
Investment Amount	BDT 107 million	BDT 200 million
IRR	**~12.9%**	**12.0%** (as given)
Strategic Fit	Core business (green tech)	**Unrelated industry**
Risk	Known and manageable	**High — lack of experience**
Opportunity Cost	Moderate (R&D sunk)	High (larger capital at risk)

Recommendation

- **Financially**, EcoCool Pro slightly outperforms the civil engineering project in IRR and offers a **positive NPV**, indicating value creation.
- **Strategically**, EcoCool Pro aligns perfectly with Mark Innovations' brand and core competencies in sustainable technology, while the alternative project does not.
- **Operationally**, EcoCool Pro leverages existing infrastructure and market positioning, unlike the alternative project which involves unfamiliar regulatory, technical, and execution risks.

Final Recommendation: **Proceed with EcoCool Pro**. It offers better strategic alignment, manageable risk, and superior financial return.

---The End---